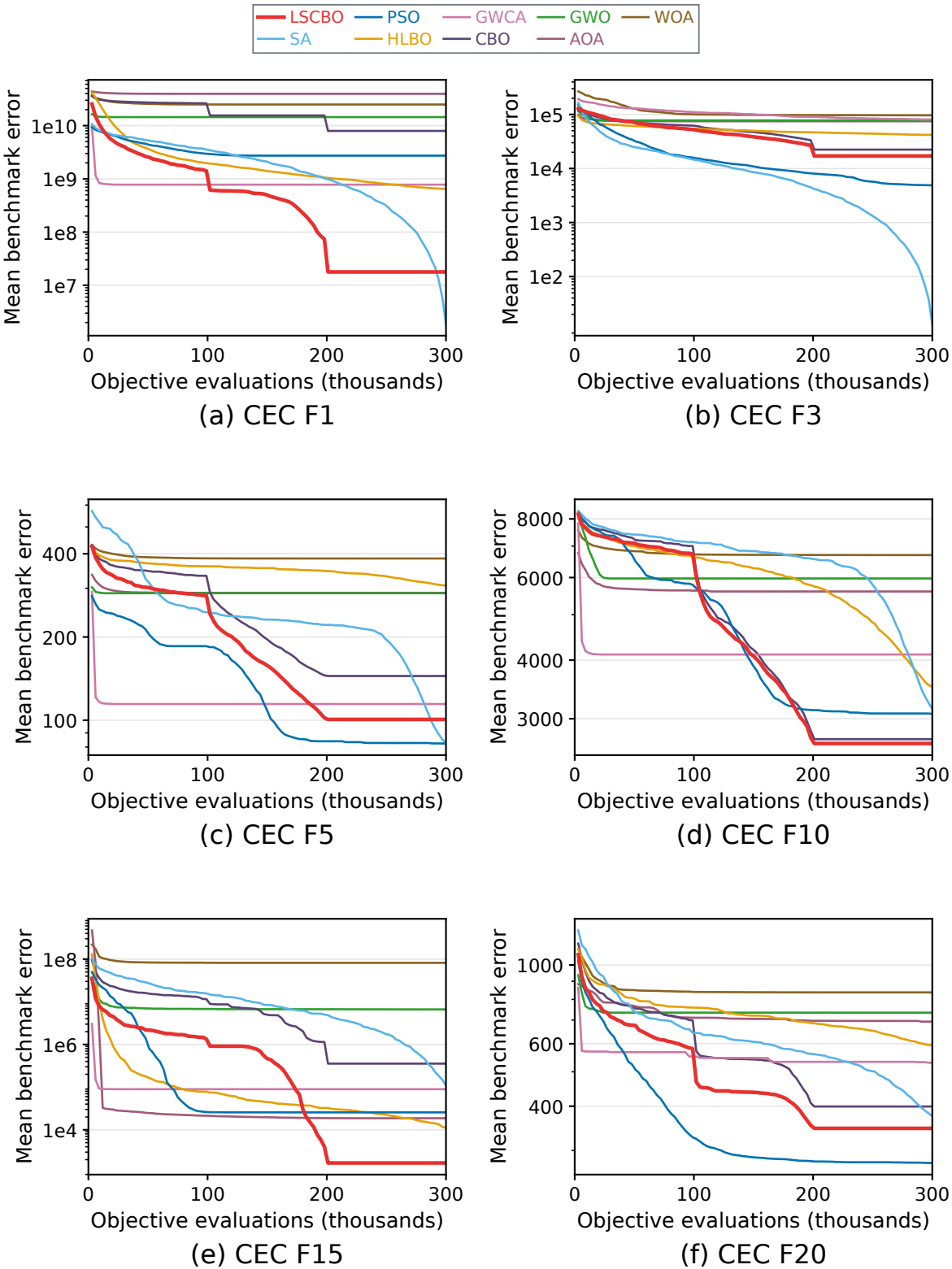
LSCBO
PSO
GWCA
GWO
WOA
SA
HLBO
CBO
AOA
Mean benchmark error
1e10
1e9
1e8
1e7
0
100
200
300
Objective evaluations (thousands)
(a) CEC F1
Mean benchmark error
1e5
1e4
1e3
1e2
0
100
200
300
Objective evaluations (thousands)
(b) CEC F3
Mean benchmark error
400
200
100
0
100
200
300
Objective evaluations (thousands)
(c) CEC F5
Mean benchmark error
8000
6000
4000
3000
0
100
200
300
Objective evaluations (thousands)
(d) CEC F10
Mean benchmark error
1e8
1e6
1e4
0
100
200
300
Objective evaluations (thousands)
(e) CEC F15
Mean benchmark error
1000
600
400
0
100
200
300
Objective evaluations (thousands)
(f) CEC F20